



INVALIDITY HELP



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The HELP facility is *not* intended to provide help with the substance of logic. You can search this document to FIND a particular word or phrase by pressing Ctrl+F.

LOGIC NOTES:

- 1) There is a brief discussion of how to approach invalidity problems in the document "Invalidity Methods" invoked by clicking on the References button.
- 2) A fairly thorough discussion of how to establish invalidity using models can be found in the TerryText in Chapter 3 sections 10 and 11. For Chapter 4, the situation is more complex, and is discussed in section 9. The technique involving infinite universe is not yet implemented in LOGIC 2010. For Chapter 5, the discussion is in sections 7 and 8, and for Chapter 6, in sections 6 and 7.

Quick Start

Use the Select button to select a problem from the program's database. The program is also able to check problems created by a User that are not in its database. To enter a User problem click the User button at the bottom of the screen.

When the problem appears in the Problem line in the upper left corner of the screen, select the size of the Universe by clicking the "click to set" button, and selecting from the drop-down list. Click the button containing a Sentence Letter, Predicate, or Name Letter to set the extension. The default extensions are: For Sentence Letters: False. For Predicates: Empty (i.e., applying to nothing). For Name Letters: 0 (i.e. naming the number 0). For n-place Operation Symbols: 0 (i.e. assigning the number 0 to every n-place sequence of individuals). For an improper description: 0 (i.e. the improper description " $\lambda x x \neq x$ ", and every other improper description, is taken as standing for the number 0). It is not possible to reset the default for the descriptive name " $\lambda x x \neq x$ ". This should not matter since every other extension can be set by the user.

You may wish to use the white workspace to copy in the argument, and do a truth functional expansion. (Note that in Chapter V, the expansion is dependent not only on the Universe, but also on the extensions of operation symbols.)

To copy the problem into the workspace, click the "Copy" button.

To do a truth functional expansion, First set a Universe size. Then, use the "Expand" button to do the expansion. In the workspace, highlight any formula or subformula beginning with a quantifier. Click the "Expand" button. The

quantifier will be eliminated in favor of a conjunction or disjunction of instances. (In the instances, the variable "i0" is taken to refer to 0, "i1" is taken to refer to 1, and so on.) In the given Universe the expanded formula is equivalent to the original. That is, the original formula is true in that Universe if and only if the resulting formula, with the quantifier eliminated, is true. When all of the quantifiers, in premises and in the conclusion, are eliminated, the problem is reduced to a problem (of invalidity) in Chapter II.

If you wish to use the facilities of the Truth Table module to determine an assignment of truth values to atomic formulas that makes the expanded premises True and the expanded conclusion False, simply highlight the expanded argument, and click the Truth Table button.

If you wish to use the Derivations module to try an Indirect Derivation EI/UI argument, hoping it will terminate without a contradiction but in a way that will indicate how to construct the model, just click the Derivation button.

After setting the extensions to produce the model you wish, click Check to check your work. If you have succeeded in providing a model in which all of the premises are true, and the conclusion is false, "Correct" will appear in the upper right hand corner of the screen. If not, the sequence of truth-values that your model *does* assign to the premises and conclusion will appear. For example, if your model makes the first premise true, the second false, and the conclusion false, you will see, "T F ∴ F". If your initial model is not successful, you may wish to make adjustments to it and try again.

Other Facilities

Print your work by clicking the Print button or pressing **Ctrl+p**. The Print window allows you to select as many problems as you wish for printing. Only the problem plus "C" (for Correct) or "I" (for Incorrect or Incomplete) will be printed. Save your work by clicking the Save button or pressing **Ctrl+s**.

Do not fear the Delete button, it always first asks you to *confirm* that you wish to delete.

You can also submit your work over the Internet to a database of homework submissions established for this course. Click the Submit button at the bottom of the screen. You will be asked to verify course information, and to enter your logic password, which you will create the first time you make a submission. You may select more than one exercise to submit. The program will return a message verifying that the submitted work has been received. You can view Assignments and your submissions by clicking the "Assignments" button on the Menu.

The Menu button brings the opening Menu module to the foreground, allowing you to select a different module or to open another instance of the present

module. The Menu module also holds access to the course website, and the Quit button for closing the program. You will be asked if you wish to save your current work before quitting.

The Close button closes the present window. You will be asked if you wish to save your current work before the window is closed.

The Feedback button takes you to a web-based channel for anonymous or open communication with the Instructor, TAs, and authors of the program. It is especially useful for comments or questions about the program, and is a source for the FAQ facility (see below). If you prefer an individual response over anonymity, include your e-mail address with your message.